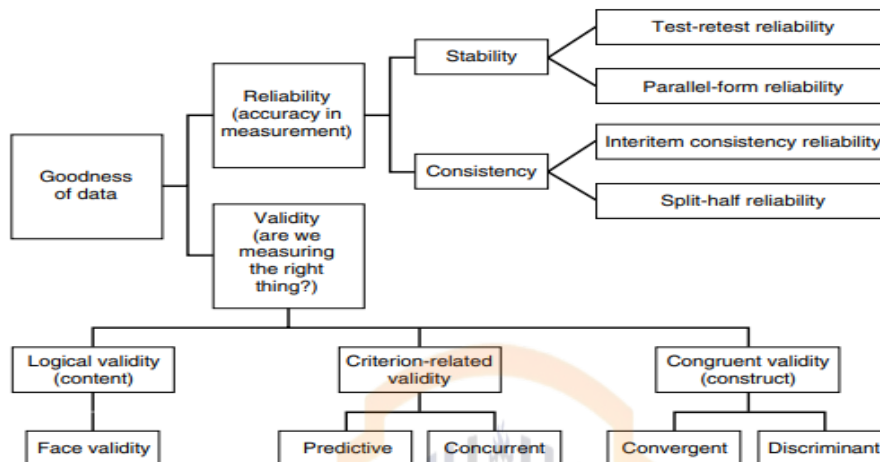


Lecture 5

Measurement: Scaling, Reliability, and Validity

Reliability is the contributor to validity. It is important to note that reliability is a necessary but not sufficient condition of the test of goodness of a measure.

Testing Goodness of Measures: Forms of Reliability and Validity.



VALIDITY

When we ask a set of questions (i.e., develop a measuring instrument) with the hope that we are tapping the concept, how can we be reasonably certain that we are indeed measuring the concept we set out to do and not something else? This can be determined by applying certain validity tests.

Several types of validity tests are used to test the goodness of measures and writers use different terms to denote them. For the sake of clarity, we may group validity tests under three broad headings:

- content validity,
- criterion-related validity, and
- construct validity.

Content Validity (surface validity): It ensures that the measure includes an adequate and representative set of items that tap the concept. The more the scale items represent the domain or universe of the concept being measured, the greater the content validity.

Face validity indicates that the items that are intended to measure a concept, do on the face of it look like they measure the concept.

Content validity: Is the test fully representative of what it aims to measure?

Face validity: Does the content of the test appear to be suitable to its aims?

Criterion validity: Do the results accurately measure the concrete outcome they are designed to measure?

Face validity requires the opinion of the layperson/test taker, while content validity is best measured by the opinion of experts.

Examples

- A test to obtain a license, such as driving or selling real estate.
- Standardized testing for academic purposes, such as the SAT and GRE.

- Tests that evaluate knowledge of subject area domains, such as biology, physics, and literature.

Criterion-Related Validity: Criterion-related validity is established when the measure differentiates individuals on a criterion it is expected to predict.

Criterion validity is made up two subcategories: predictive and concurrent. Predictive validity refers to the extent to which a survey measure forecasts future performance.

Concurrent validity is established when the scale discriminates individuals who are known to be different; that is, they should score differently on the instrument. To have concurrent validity, an indicator must be associated with a preexisting indicator that is judged to be valid.

For example: we create a new test to measure intelligence. For it to be concurrently valid, it should be highly associated with existing IQ tests (assuming the same definition of intelligence is used). It means that most people who score high on the old measure should also score high on the new one, and vice versa. The two measures may not be perfectly associated, but if they measure the same or a similar construct, it is logical for them to yield similar results.

Predictive validity: Criterion validity whereby an indicator predicts future events that are logically related to a construct is called a predictive validity. It cannot be used for all measures. The measure and the action predicted must be distinct from but indicate the same construct.

Predictive measurement validity should not be confused with prediction in hypothesis testing, where one variable predicts a different variable in future.

For example: Look at the scholastic assessment tests being given to candidates seeking admission in different subjects. These are supposed to measure the scholastic aptitude of the candidates – the ability to perform in institution as well as in the subject.

If this test has high predictive validity, then candidates who get high test score will subsequently do well in their subjects.

If students with high scores perform the same as students with average or low score, then the test has low predictive validity.

Construct Validity: Construct validity is for measures with multiple indicators. This concept has two related parts: A construct is an idea that has related parts based on a theory but is not a physical thing. Depression, anxiety and stress are all psychological constructs.

A good example of construct validity is a test that accurately measures mathematical reasoning skills, correlates highly with other established math tests, and predicts performance in math-related tasks.

It addresses the question:

If the measure is valid, do the various indicators operate in consistent manner?

It requires a definition with clearly specified conceptual boundaries. In order to evaluate construct validity, we consider both theory and the measuring instrument being used.

This is assessed through convergent validity and discriminant validity.

Convergent Validity This kind of validity applies when multiple indicators converge or are associated with one another. Convergent validity means that multiple measures of the same construct hang together or operate in similar ways.

For example: we construct “education” by asking people how much education they have completed, looking at their institutional records, and asking people to complete a test of school level knowledge. If the measures do not converge (i.e. people who claim to have college degree but have no record of attending college, or those with college degree perform no better than high school dropouts on the test), then our test has weak convergent validity and we should not

combine all three indicators into one measure.

Discriminant Validity (divergent validity): Discriminant validity is the opposite of convergent validity. It means that the indicators of one construct hang together or converge, but also diverge or are negatively associated with opposing constructs. It says that if two constructs A and B are very different, then measures of A and B should not be associated.

For example: we have 10 items that measure political conservatism. People answer all 10 in similar ways. But we have also put 5 questions in the same questionnaire that measure political liberalism.

Our measure of conservatism has discriminant validity if the 10 conservatism items hang together and are negatively associated with 5 liberalism ones.

